

An Inversion-Preserving Bijection Between Cayley Permutations and Fubini Rankings

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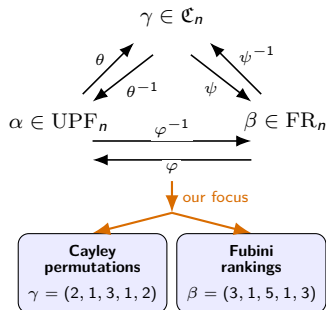
Outline

- ▶ Motivation and definitions
- ▶ Question
- ▶ Result

Motivation

- We studied three families. Our focus here will be on Cayley permutations and Fubini rankings.

Three families in the paper



Cayley Permutations

A **Cayley permutation** of length n is a tuple

$$\gamma = (c_1, \dots, c_n) \in [n]^n$$

whose distinct values are $1, 2, \dots, m$ for some $m \leq n$.

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There can be no missing values before the largest one.

Repeated values are allowed.

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$$\gamma = (2, 1, 3, 1, 2)$$

The values that appear are $\{1, 2, 3\}$.

No values are missing, so
 γ is a Cayley permutation.

Cayley Permutation Check

Are the following Cayley permutations?

1. $(1, 1, 1, 1, 1)$

Cayley Permutation Check

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Yes

Cayley Permutation Check

Are the following Cayley permutations?

1. $(1, 1, 1, 1, 1)$
2. $(4, 2, 1, 3, 2)$

Yes

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Yes

Cayley Permutation Check

Are the following Cayley permutations?

1. $(1, 1, 1, 1, 1)$

Yes

2. $(4, 2, 1, 3, 2)$

Yes

3. $(3, 1, 3, 1, 3)$

Cayley Permutation Check

Are the following Cayley permutations?

- | | |
|----------------------|-----|
| 1. $(1, 1, 1, 1, 1)$ | Yes |
| 2. $(4, 2, 1, 3, 2)$ | Yes |
| 3. $(3, 1, 3, 1, 3)$ | No |

It is not a Cayley permutation because the value 2 is missing.

Fubini Rankings

A **Fubini ranking** of length n is a tuple

$$\beta = (\beta_1, \dots, \beta_n) \in [n]^n$$

in which rank 1 appears, and if a rank i occurs k times, then the next larger rank that appears is $i + k$.

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If k competitors tie at rank i , then the next rank is $i + k$.

Ties(repeats) are allowed.

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Ties(repeats) are allowed.

$$\beta = (3, 1, 5, 1, 3)$$

Rank 1 occurs twice, so the next rank is $1 + 2 = 3$.

Rank 3 occurs twice, so the next rank is $3 + 2 = 5$.

Therefore, β is a Fubini ranking.

Fubini Ranking Check

Are the following Fubini rankings?

1. $(1, 2, 3, 4, 5)$

Fubini Ranking Check

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Fubini Ranking Check

Are the following Fubini rankings?

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| 1. $(1, 2, 3, 4, 5)$ | Yes |
| 2. $(1, 1, 3, 3, 5)$ | Yes |
| 3. $(1, 1, 1, 3, 5)$ | No |

It is not a Fubini ranking because after three 1's, the next rank must be 4.

Our title is

An **Inversion** -Preserving **Bijection**
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Bijections

A function $f: A \rightarrow B$ is a **bijection** if it is both *one-to-one* and *onto*.

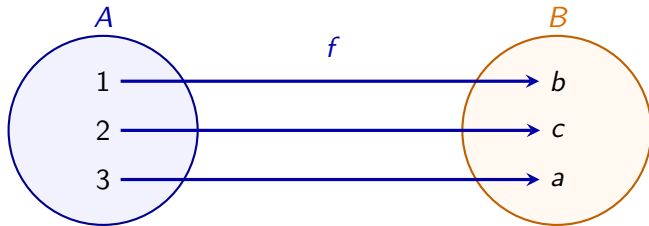
In words: Every object in B is paired with *exactly one* object in A .

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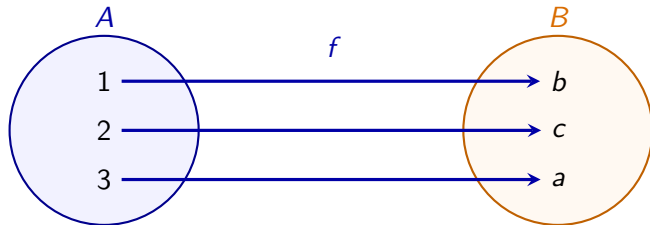


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Example



No two inputs share an output, and no output is missed.

Therefore, f is a bijection and $|A| = |B|$.

Inversions

Let $w = (w_1, \dots, w_n)$. An **inversion** is a pair of positions (i, j) such that $i < j$ but $w_i > w_j$.

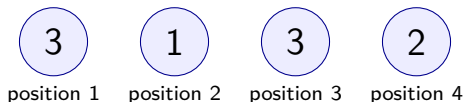
In words: A larger entry appears before a smaller entry.

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Example: $w = (3, 1, 3, 2)$

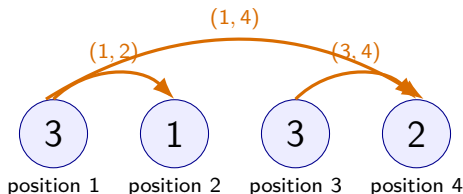


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Example: $w = (3, 1, 3, 2)$



Each arrow goes from a larger entry to a smaller entry on its right.

$$\text{Inv}(w) = \{(1, 2), (1, 4), (3, 4)\}, \quad \text{inv}(w) = 3.$$

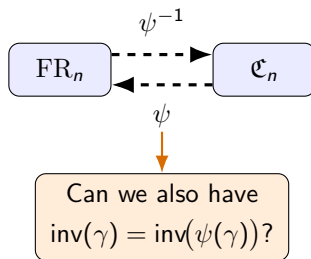
The two equal entries 3 and 3 do not form an inversion.

Motivation

- ▶ We studied three families. Our focus here will be on Cayley permutations and Fubini rankings.
- ▶ Can we construct a bijection

$$\psi : \mathfrak{C}_n \longrightarrow \text{FR}_n$$

that preserves inversions?



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Suppose $\gamma \in \mathfrak{C}_n$ has k inversions.



In words: Applying ψ may change the tuple, but it does not change how many pairs are out of order.

$$\text{inv}(\gamma) = \text{inv}(\psi(\gamma)).$$

Main Result

Theorem

For any $n \in \mathbb{N}$, the sets $\mathfrak{C}_n$ and FR_n are in bijection.

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Lemma 1

Given $\gamma = (c_1, c_2, \dots, c_n) \in \mathfrak{C}_n$, define $g : \mathfrak{C}_n \rightarrow \text{FR}_n$ by $g(\gamma) = (b_1, b_2, \dots, b_n)$, where $b_i = 1 + |\{1 \leq j \leq n : c_j < c_i\}|$. for every $i \in [n]$. The map g is well defined.

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Lemma 2

Given $\beta = (b_1, b_2, \dots, b_n) \in \text{FR}_n$, let $r_1, r_2, \dots, r_m$ be the $m \leq n$ distinct ranks appearing in β , where $r_1 < r_2 < \dots < r_m$. Define $\psi : \text{FR}_n \rightarrow \mathfrak{C}_n$ by $\psi(\beta) = (c_1, c_2, \dots, c_n)$, where $c_i = j$ whenever $b_i = r_j$. The map ψ is well defined.

$\mathfrak{C}_n$ to FR_n mapping

Start with a Cayley permutation

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Count the entries in the **whole tuple** that are strictly smaller than c_i . Then add 1.

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$$(\gamma)_i = \psi(\gamma)_i = 1 + |\{j \in [n] : c_j < c_i\}|$$

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position i	1	2	3	4	5
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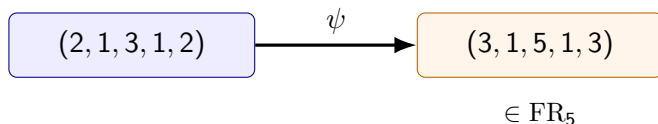
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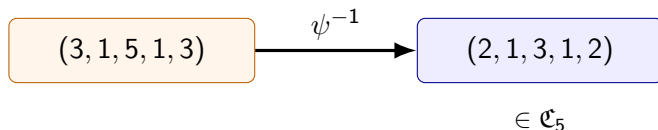
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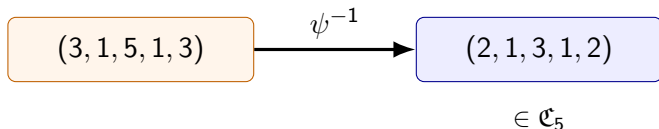
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This undoes the other mapping, returning the Cayley permutation we started with.

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Theorem

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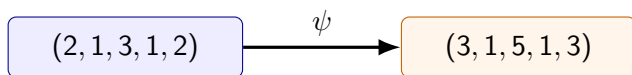
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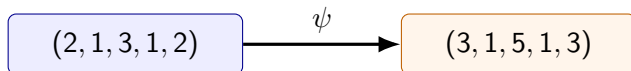
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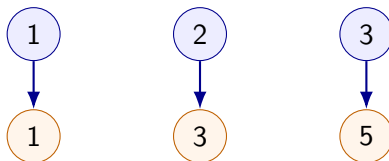
How Are Inversions Preserved?



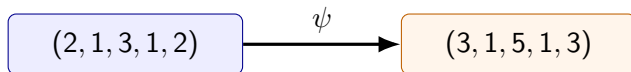
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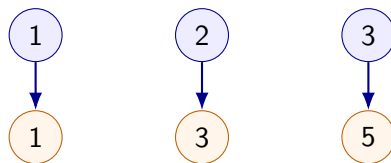
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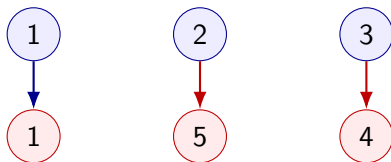
The values change like this:



Order is **preserved**: $1 < 2 < 3$ becomes $1 < 3 < 5$.

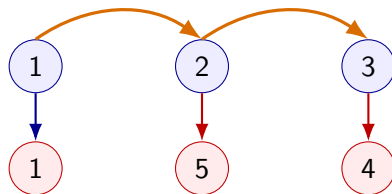
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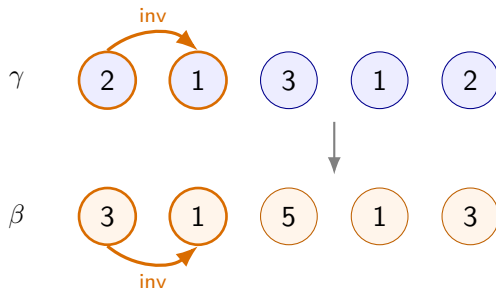


$1 < 5$, but not $5 < 4$

Order fails, so inversions would not be preserved.

How Are Inversions Preserved?

Returning to our map, look at positions (1, 2):



$2 > 1$ becomes $3 > 1$, still an inversion.

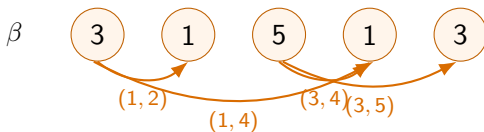
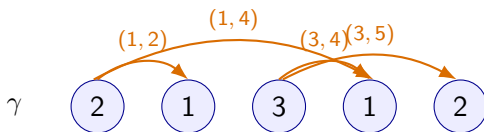
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Both tuples have the **same** inversion pairs:



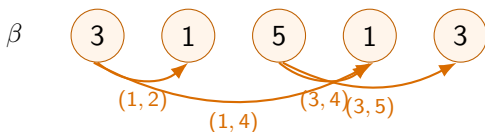
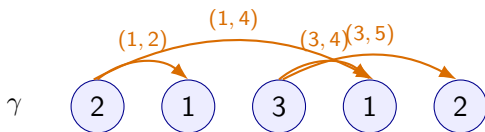
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$$\text{Inv} = \{(1, 2), (1, 4), (3, 4), (3, 5)\}, \quad \text{inv} = 4.$$

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$$1 + \#\{\text{distinct ranks smaller than } b_i\}.$$

- ▶ The relabeling preserves every comparison, so

$$\text{Inv}(\gamma) = \text{Inv}(\psi(\gamma)).$$

Summary

We constructed an explicit **inversion-preserving bijection** between Cayley permutations and Fubini rankings.

- ▶ The map ψ replaces each c_i by

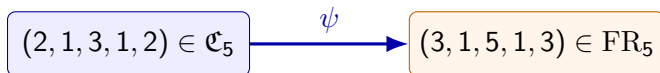
$$1 + \#\{\text{entries smaller than } c_i\}.$$

- ▶ The map ψ^{-1} replaces each b_i by

$$1 + \#\{\text{distinct ranks smaller than } b_i\}.$$

- ▶ The relabeling preserves every comparison, so

$$\text{Inv}(\gamma) = \text{Inv}(\psi(\gamma)).$$



inv = 4 on both sides

Thank You!



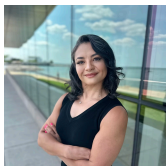
Email: vo.tung@stthom.edu

Website: tung-vo.com

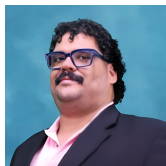
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